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Cooperative or competitive? Resolving social dilemmas in autonomous vehicles through evolutionary game theory

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ABSTRACT

With the large-scale deployment of autonomous vehicles (AVs), AV-human-driven vehicle (HV) interactions are increasingly common. AVs face a social dilemma: competitive behavior raises ethical and public acceptance concerns, whereas cooperative behavior can invite exploitation and degrade efficiency. We use Evolutionary Game Theory (EGT) to model long-run adaptation between AVs and HVs and quantify agent sociality via a data-calibrated Social Value Orientation (SVO) metric. After calibrating HV social preferences from unprotected left-turn trajectories, we incorporate HV heterogeneity into a two-population EGT with cooperative and competitive types. SVO-informed rewards are used to construct payoff matrices for replicator analyses to identify evolutionarily stable strategies (ESS). Experiments show that AV policies with moderate egoism mitigate the social dilemma and tend to achieve population-level dominance in both roles (left-turning and straight-going), whereas overly cooperative policies are evolutionarily unstable. Moreover, AVs benefit from opponent-aware, dynamically adjustable sociality to accommodate diverse HV preferences. To test the theory, we run agent-based imitation simulations. Sensitivity analyses indicate that AV advantages are hard to observe at low market penetration but become pronounced as penetration approaches about 50%, after which convergence accelerates. Overall, the framework clarifies when and why AV sociality preferences succeed over time, offering actionable guidance for designing adaptive, socially compatible AV decision policies in mixed traffic.

1. Introduction

With breakthroughs in AI and deep learning, human-robot interaction is shifting from tool-based to partnership-based; yet aligning machine behavior with human social norms remains challenging (Gu et al., 2017, Awad et al., 2018, Soori et al., 2023, Torras 2024). On public roads, this tension manifests in mixed traffic, where autonomous vehicles (AVs) share the road with human-driven vehicles (HVs), making their strategic choices deeply interdependent.

In the current stage of human-robot coexistence, humans' natural self-interest poses a challenge to robot behaviors (Rodríguez-Guerra et al., 2021). AVs often default to conservative policies for safety, which can be exploited by HVs (e.g., opportunistic cutting-in), degrading traffic efficiency and user acceptance (Bonnefon et al., 2016, Shalev-Shwartz and Shammah, 2016).

This trade-off constitutes a social dilemma, as illustrated in Fig. 1: cooperative AV policies can penalize performance, whereas competitive policies raise ethical concerns and public distrust; both can trigger defensive human reactions, reduce flow efficiency and perpetuate a self-defeating cycle that compromises safety and adoption (Mertens et al., 2020, Huang et al., 2023, Zhang et al., 2024).

Crucially, mixed traffic is persistent, and intersections serve as repeated venues for ubiquitous yet one-shot HV-AV encounters. Although each encounter is transient, population composition and AV policy orientation shape long-run outcomes. Addressing this problem is a marathon, not a sprint, requiring population-level adaptation rather than single-interaction optimization. Moreover, HVs are heterogeneous in risk appetite and willingness to cooperate; ignoring this diversity obscures how population preferences drive macroscopic efficiency and safety.

Intersections are prototypical high-interaction sites and account for about 40% of traffic crashes (Administration 2008), and unprotected left-turn is the most representative case (Najm and Toma, 2013, Sun et al., 2020, Tian et al., 2025). In this scenario, even Waymo's AVs, which are among the most advanced in the AV field, exhibit overly conservative behaviors (Noh 2019). These overly conservative behaviors not only reduce the efficiency of AVs but may also frustrate other drivers, leading to road rage, deadlocks, and traffic accidents (D'Onfro

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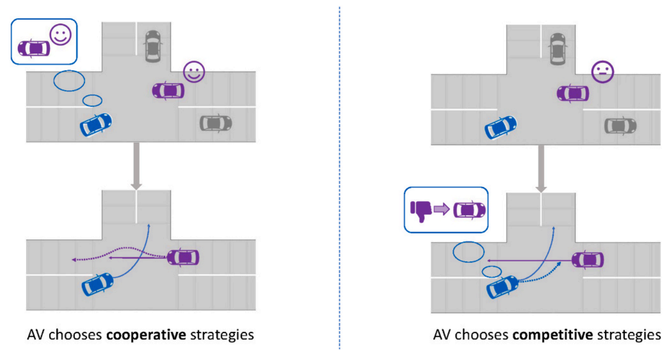

Fig. 1. An example of a social dilemma. **Left**: When AV (in purple) adopts a cooperative strategy, HV (in blue) takes advantage of the AV's cooperative tendency and speeds through, forcing the AV to yield. **Right**: When AV adopts a competitive strategy to prioritize its own benefit, it disrupts the behavior of other vehicles, reflecting potential moral condemnation that AVs may face. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2018), representing a classic example of a social dilemma.

In summary, the resolution of social dilemmas faced by AVs during interactions—and the mechanisms governing their long-term evolution—remain insufficiently understood. Two gaps remain in explaining AV social dilemmas:

- Long-term interaction dynamics. Most existing studies evaluate AV policies at the scenario level or over short horizons, without explicitly depicting how AV–HV interaction patterns evolve over extended periods as users adapt.
- Heterogeneous human drivers. Many models treat HVs as a homogeneous population or rely on an "average" driver, overlooking how heterogeneous social preferences (cooperative vs. competitive types) influence long-run interaction trajectories.

To capture long-run, strategic adaptation with heterogeneous drivers, we adopt an EGT framework: it models population learning and adaptation under bounded rationality and anonymous matching, and has been widely used to analyze strategy adaptation, behavioral dynamics, and population outcomes (Weibull 1997, Abd et al., 2019, Awaga et al., 2020). The core research question is: How should AV sociality preferences be designed in the long run, given heterogeneous HV populations?

Building on this perspective, we develop a two-population evolutionary game-theoretic model for the unprotected left-turn scenario, leveraging data-calibrated sociality preferences to assess how AV social choices shape long-run outcomes in mixed traffic. The findings provide actionable guidance for selecting dynamic, adaptive AV sociality policies in mixed traffic.

The main contribution of this study is a population-level analysis of how alternative AV sociality preferences affect long-run performance in mixed traffic, thereby clarifying the social dilemmas AVs may encounter and offering practical design insights. We further address two modeling challenges:

- Long-term interaction. We construct a two-population evolutionary game between LT and SG roles and analyze the resulting equilibria, basins of attraction, and uncertainty bands, thereby depicting the long-run AV–HV interaction process rather than only single episodes.
- Heterogeneity-aware strategy. We incorporate heterogeneous human drivers by classifying HVs into cooperative and competitive types based on calibrated SVO, and we examine how AV strategies should adjust their sociality preference under different HV population compositions.

The structure of the rest of the paper is as follows: In Section 2, we review relevant research on social interactions in autonomous driving and evolutionary game theory. In Section 3, we present the construction of the evolutionary game theory model, including problem description, modeling assumptions, and the payoff matrix. Section 4 presents the evolutionary analysis, uncertainty bands, and sensitivity to AV penetration and preference concentration. Finally, Section 5 summarizes the paper, highlighting the insights gained from the study and suggesting directions for future research.

2. Literature review

Given the long-term, population-level nature of AV–HV interactions and the social dilemmas arising from cooperative vs. competitive tendencies, we review two key areas closely related to these issues and the latest research developments: (1) social interaction for AVs, and (2) evolutionary game theory.

2.1. Social interaction for AVs

The interaction in autonomous driving refers to the interplay between AV and other road users (HVs, AVs, pedestrians). It can be defined as "the behavior of at least two road users that can be interpreted as being influenced by the potential for them to simultaneously occupy the same spatial area in the near future." (Wang et al. 2022), and we focus on AV–human interactions in mixed traffic.

From a methodological perspective, decision making for AVs in interactive settings can be broadly grouped into learning-based and model-based approaches. Learning-based methods—including imitation learning, deep reinforcement learning, and inverse reinforcement learning—learn interaction policies directly from data, capturing rich, context-dependent patterns and uncertainty (Gupta et al., 2018, Kiran et al., 2022, Valiente et al., 2022, Hang et al., 2023, Crosato et al., 2024, Liu 2024).

Model-based methods encode structural assumptions about vehicle dynamics and interaction and can be further divided into three main families. Rule-based and gap-acceptance models implement handcrafted heuristics for lane changing, merging, and yielding, typically using thresholds on headways, time-to-collision, or priority rules; such models remain widely used as baselines and background-traffic generators. Optimization- and MPC-based motion planners formulate interaction as a constrained optimal control problem, where the AV minimizes a cost on path tracking, comfort, and safety while reacting to predicted trajectories of surrounding vehicles (Li et al., 2018, Ji and Levinson 2020). Game-theoretic and interactive-planning approaches explicitly reason about mutual influence and strategic behavior, computing Nash or Stackelberg equilibria or receding-horizon best responses to generate socially compliant trajectories (Fisac et al., 2019, Sarkar and Czarnecki 2021). Scenario-based testing and assessment frameworks integrate these controllers into systematic environments to probe safety margins under interactive driving and connect microscopic decisions to macroscopic safety performance (Sun et al., 2018, Sun et al., 2022, Hang et al., 2023, Zhang et al. 2024, Wei et al., 2025, Zhou et al., 2025).

In mixed traffic, AVs must anticipate human drivers whose behavior is heterogeneous and context-dependent, which challenges both prediction and mutual understanding (Sun et al., 2018, Schwarting et al., 2019, Hang et al., 2023). Beyond algorithmic modeling, human-factors evidence documents systematic variability in risk attitude, cooperation, and attention—factors that shape interaction outcomes(Liu 2024). However, merely reproducing observed behavior or optimizing short-term performance in a fixed scenario is insufficient for social adaptability: AVs must reason about trade-offs between self-interest and others' welfare at the long-run composition of behaviors (equilibria and basins), not only single encounters (Sun et al., 2018, Markkula et al., 2020).

A social-interaction perspective frames driving behavior as a trade-off between self-interested and other-regarding motives. It specifies how agents make context-sensitive compromises so that actions align with social norms and collective welfare. For AVs, this perspective offers a principled way to infer and execute socially appropriate behavior (Adnan et al., 2018).

Social Value Orientation (SVO), originating in social psychology and decision science, quantifies how much weight an individual places on others' outcomes relative to their own (Liebrand 1984). SVO has been extensively applied across social psychology, behavioral economics, and game theory.

In autonomous driving, SVO is used to characterize drivers along a cooperation–competition spectrum. These social preferences can be inferred from driving trajectories via inverse reinforcement learning (IRL) and incorporated into AV decision-making and interaction policies (Liu et al., 2024). Schwarting et al. integrated SVO into AV systems to promote socially compatible interactions with human drivers (Schwarting et al., 2019). Building on this idea, subsequent studies embed SVO in prediction and planning modules, improving interaction smoothness and enabling human-like yet safety-preserving decisions (Schwarting et al., 2019, Le and Malikopoulos 2022, Liu et al., 2024). While real-time SVO estimation faces practical challenges (compute budget, noise, cross-cultural transfer), offline calibration coupled with population analysis offers a tractable path for understanding long-run social adaptability.

Overall, SVO provides an effective method for quantifying the social preference of drivers, thereby facilitating the optimization of decision-making processes in autonomous driving systems to achieve more efficient interactions within the autonomous driving framework. Although integrating SVO into autonomous driving systems can improve the accuracy of human trajectory prediction, this approach still faces certain limitations in practical applications, such as: how to estimate the SVO of surrounding vehicles in real-time without compromising decision-making efficiency; inability to fully capture the influence of complex factors like emotions and fatigue on driving behavior; and how to generalize the model across different cultural contexts and traffic scenarios.

2.2. Evolutionary game theory

EGT models how strategy shares evolve when agents imitate or adopt higher-payoff behaviors under bounded rationality and anonymous matching (Weibull 1997). It emphasizes stability concepts such as ESS (Kasthurirathna et al., 2015) which refers to a strategy that, once adopted by the majority of a population, cannot be easily invaded or replaced by alternative strategies. ESS refines the Nash equilibrium by introducing a robustness condition that makes it resistant to small perturbations (Smith and Price 1973). The development of EGT has provided crucial theoretical tools for understanding biological behaviors, the dynamics of competition and cooperation within social groups, and the evolution of human behavior. EGT has garnered increasing interest and has been widely applied across various fields, including economics, sociology and anthropology (Bauch and Bhattacharyya 2012, Conroy-Beam et al., 2015, Li et al., 2024).

EGT has been applied in transportation to study behavioral interactions, control, routing, and safety (Ahmad et al., 2023), which aligns with the collective and adaptive nature of traffic systems. Given traffic's collective, adaptive nature, EGT provides a natural modeling lens for transportation interactions (Shivshankar and Jamalipour 2015, Ji and Levinson 2020). EGT is well-suited to capture these characteristics, providing a robust modeling approach to represent and analyze such interactions within transportation systems.

With the emergence of AVs, they have gradually become an integral part of road networks, influencing the dynamics of these networks. Scholars have increasingly applied EGT to analyze the changes introduced by AVs within transportation systems. Research has explored the factors and mechanisms governing interactions between AVs and pedestrians, validating the traditional pedestrian priority rules and the rationale behind collision-warning systems, with the aim of further promoting the adoption of autonomous driving technologies (Rui et al., 2023). Li and Sun utilized a three-party evolutionary game model to investigate the service mode choices among passengers, traditional taxis, and autonomous taxis, highlighting the significant influence of autonomous taxi fare structures, fleet size, and technological advancements on the development of this sector (Li and Sun 2024).

Existing research on autonomous driving using evolutionary game theory primarily concentrates on the impact of AV integration within traffic systems. However, such studies often overlook the strategic adaptation and long-term evolution of AVs themselves. They rarely employ evolutionary game models as tools to examine the broader developmental trajectory of autonomous driving technology.

As autonomous driving technology advances, issues related to ethical decision-making and social responsibility have become increasingly prominent. When faced with social dilemmas, excessive competitiveness may exacerbate potential traffic conflicts and accidents, while excessive cooperation could encourage human drivers' "bullying" behavior, thereby intensifying such inequities.

We integrate data-calibrated SVO into a two-population EGT for unprotected left-turn, and we analyze sensitivity to AV penetration (AV share by role) and preference polarization (dispersion of SVO-based preferences). We further report uncertainty bands for equilibria and basins, yielding robust design intervals for AV sociality. Under weak selection, discrete Fermi dynamics yield the same direction and fixed points as the replicator; we later verify this via an individual-based simulation.

3. Problem modeling

In this section, we set up an asymmetric two-population evolutionary game in the unprotected left-turn scenario, with the left-turning (LT) and straight-going (SG) flows as the two populations and AV and HV as the two strategy types in each population. This formulation allows us to examine how different LT–SG strategy combinations shape population stability in unprotected left-turn. To parameterize payoffs, we introduce SVO-based preference weights estimated from trajectories; on this basis we derive the conditions for ESS.

3.1. Basic model formulation and assumptions

We consider an unprotected left-turn intersection where left-turning and straight-going vehicles constitute the two interacting populations. In each interaction, one vehicle is drawn at random from LT and one from SG; each interaction is a simultaneous-move, incomplete-information game. Each vehicle adopts either HV (human-driven) or AV (automated control policy) in its respective role.

The observed performance of each type of vehicle during interactions feeds back into adoption decisions—i.e., whether individuals choose AVs or HVs. This mirrors the real world: consistently superior interaction performance by AVs can increase AV uptake (reducing HV share), whereas superior HV performance can shift adoption in the opposite direction.

Based on the above modeling content, the following assumptions are made:

Assumption 1: Left-turning vehicles and straight-going vehicles lack knowledge of each other's role decisions during each interaction.

Assumption 2: Relative performance affects market composition: better AV (resp. HV) outcomes attract more users, altering population shares.

Assumption 3: The interaction process may or may not converge to an equilibrium. At equilibrium, the two populations maintain stable preferences over AV/HV unless perturbed by exogenous changes (e.g., technology or infrastructure).

The parameters used in this study and their definitions are listed in Table 1.

3.2. Svo-based evolutionary game theory

3.2.1. Interactions in unprotected left-turn intersection

Sociality preference reflects how interacting vehicles balance self-interest and regard for others. High-interaction scenarios are thus informative for quantifying sociality preference; unprotected left-turn intersection is a canonical example. As shown in Fig. 2, the LT and SG trajectories intersect at a conflict point, requiring both parties to adjust speed and position within limited time and space to avoid collision.

In this game-like situation, the LT vehicle must assess the SG vehicle's speed and distance to decide whether to turn now or wait, while the SG vehicle infers the LT vehicle's intent and decides whether to maintain speed, yield, or brake. This dynamic interaction underscores the need for both parties to adjust strategies rapidly to achieve safe and efficient passage.

Following Liu et al. (2024), we model the coordination between the LT and SG vehicles as an iterated best-response (IBR) process in a dynamic game. At each decision step, both vehicles are assumed to observe the current traffic state through their perception module, including positions, velocities, lanes and gap geometry of the interacting pair and nearby background vehicles. Given this observed state, one vehicle (LT or SG) takes the other's current planned trajectory as a short-term prediction of its intention and computes its own response by solving an optimal-control subproblem under the reward function defined in Section 3.2.2, which trades off safety, efficiency and conflict resolution according to the calibrated sociality weights. The opponent then updates its plan by solving the same type of best-response problem with the first vehicle's new trajectory held fixed. This alternating procedure repeats until the joint trajectories converge or a maximum number of iterations is reached. In this formulation, "coordination" (i.e., which vehicle yields and which passes first) is not hard-coded but emerges as the equilibrium outcome of the IBR game, while the observe–predict–respond cycle maps naturally onto the perception–prediction–planning pipeline of real-world AV controllers.

3.2.2. Social value orientation and reward function modeling

To characterize social preference during interactions and incorporate social factors into payoffs, we build on Liu et al. (Liu et al., 2024), introducing a calibration model for social preference and an SVO-aware reward function.

The reward function is a fundamental component of the game model. To demonstrate the social aspect through the reward function, we define two individual rewards—driving process R_{I1}, lane deviation R_{I2} and one group reward—conflict resolution R_G. The overall reward function is a linear combination of the three terms, as shown in Eq. (1).

$$U = w_{I1} \times R_{I1} + w_{I2} \times R_{I2} + w_G \times R_G \tag{1}$$

where w_{I1}, w_{I2} and w_G are the weight to be calibrated. These weights specify how much the driver values driving process, lane deviation, and conflict resolution, respectively. To avoid scale ambiguity and improve numerical stability, we constrain the weight vector $w = (w_{I1}, w_{I2}, w_G)$ to lie in a bounded, unit-norm set:

$$-1 < w_{I1}, w_{I2}, w_G < 1, w_{I1}^2 + w_{I2}^2 + w_G^2 = 1.$$

After each update, the vector w is projected back onto this unit sphere so that only the relative importance between the three components, rather than their absolute scale, influences behavior.

To calibrate the sociality weights w from naturalistic LT–SG trajectories, we adopt the game-theoretic inverse reinforcement learning (GT-IRL) algorithm proposed by Liu et al. (Liu et al. 2024), rather than a maximum-entropy IRL formulation. The stage reward in (1) can be written as

$$r_\theta\left(s_t, a_t^{LT}, a_t^{SG}\right) = \theta^\top \mu\left(s_t, a_t^{LT}, a_t^{SG}\right) \tag{2}$$

where $\mu(\cdot)$ collects the driving process, lane deviation and conflict resolution features and $\theta = (w_{I1}, w_{I2}, w_G)$ is the reward parameter vector to be calibrated. From the demonstrative trajectories, we first compute the empirical feature expectation vector μ_E. GT-IRL then iteratively alternates between:(i) game-theoretic simulation, where an iterated best-response (IBR) solver computes equilibrium joint trajectories and their feature expectations under the current θ; and (ii) reward update, where θ is updated by solving a max–min feature-matching problem that finds a reward direction maximising the minimum margin between μ_E and the feature expectations of the generated policies. The procedure stops when this margin falls below a small threshold, and the resulting θ^* is taken as the calibrated reward parameter vector and subsequently mapped to an SVO angle φ.

To explicitly model social preference within this framework, a weighting parameter φ is introduced and formally defined in Eq. (3).

$$\varphi = \operatorname{argtan}\left(\frac{w_G}{\sqrt{(w_{I1}{}^2 + w_{I2}{}^2)}}\right) \tag{3}$$

which captures the balance between group interest and self-interest. Following the SVO ring paradigm in social psychology and its use in AV interaction studies (e.g., (Schwarting et al., 2019, Liu et al., 2024)),

Table 1
Notations.

Parameters	Definitions
φ	sociality preference
U	reward function
R_{I1}	driving process
R_{I2}	lane deviation
R_G	conflict resolution
w_{I1}	weight of driving process
$w_{I2} w_G$	weight of lane deviation weight of conflict resolution

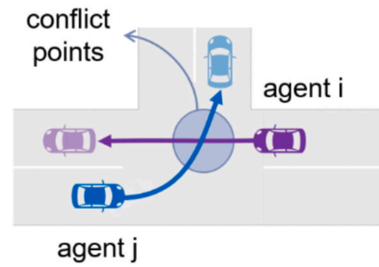

Fig. 2. Left-turning vehicle and straight-going vehicle in unprotected left-turn intersection.

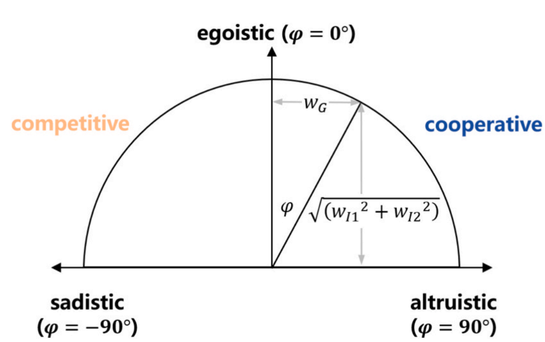

Fig. 3. The definition of sociality preference. Angles near 90° indicate altruistic preferences, angles near − 90° indicate sadistic preferences, and angles near 0° indicate egoistic preferences.

we interpret the SVO angle φ as shown in Fig. 3. When the angle approaches 90°, it indicates that the driver exhibits a strong tendency toward cooperation, reflecting an absolute altruistic orientation. Conversely, when the angle approaches − 90°, it signifies that the driver demonstrates strongly competitive (spiteful) orientation, prioritizing self-interest over cooperative interactions. Additionally, when the angle nears 0°, it represents a purely self-interested (egoistic) social preference, where the driver's decision-making is primarily guided by individual utility maximization. In our reward parameterization, φ is directly linked to the group-oriented weight w_G: $\varphi > 0$ corresponds to $w_G > 0$, meaning that group-level outcomes enter the reward positively, whereas $\varphi < 0$ corresponds to $w_G < 0$, indicating that group outcomes are not valued or can even be sacrificed for self-interest. Therefore, using $\varphi = 0^\circ$ as the boundary between cooperative and competitive regimes is both theoretically consistent with the SVO literature and aligned with our reward specification.

To clarify the overall methodology and the connection between naturalistic data, IRL-based sociality calibration, and evolutionary game analysis, Fig. 4 provides an overview of the proposed data-to-game modeling pipeline. The three steps correspond to interaction extraction and reward definition, GT-IRL calibration, and mapping calibrated preferences to SVO-based payoff construction.

3.2.3. *Evolutionary game theory model*

To study how AVs with different sociality preferences influence long-run traffic dynamics in mixed driving, we develop a two-population evolutionary game in unprotected left-turn intersection. LT and SG are modeled as two interacting populations; each population's members choose between HV and AV, forming a two-strategy evolutionary game in each population. Although a vehicle can either turn left or go straight, LT and SG are treated as distinct populations at intersections due to their different priorities and decision structures (Zhang et al., 2025).

Let $p_{LT} \in [0,1]$ denote the HV proportion among LT vehicles, so the AV proportion is $1 - p_{LT}$. Here, $p_{LT} = 0$ indicates that there are no HVs willing to be used among left-turning vehicles, and $p_{LT} = 1$ indicates that all left-turning vehicles prefer to use HVs. Similarly, let $p_{SG} \in [0,1]$ be the HV proportion among SG vehicles, with AV share $1 - p_{SG}$. $p_{SG} = 0$ means that there are no HVs willing to be used among straight-going vehicles, and $p_{SG} = 1$ means that all straight-going vehicles prefer to use HVs. The values of p_{LT} and p_{SG} are flexible and can be tailored according to the specific circumstances of the real world. They can be viewed as the market penetration rates of AVs and HVs.

Based on these definitions, we next construct the payoff matrices and replicator dynamic equations for both left-turning and straight-going vehicles, and analyze their strategic evolution.

To clarify how the calibrated reward function is used to build the payoff matrices, we briefly summarize the procedure.

(1) From naturalistic encounters to type-level weights. For each naturalistic LT–SG encounter n, the GT-IRL procedure in Section 3.2.2 yields a calibrated weight vector

$$\theta^{(n)} = \left(w_{I1}^{(n)}, w_{I2}^{(n)}, w_G^{(n)}\right) \tag{4}$$

Each $\theta^{(n)}$ is mapped to an SVO angle $\phi^{(n)}$, and the corresponding human driver is classified as cooperative if $\varphi^{(n)} > 0^\circ$ and competitive if $\varphi^{(n)} < 0^\circ$. For each HV type $\beta \in \{\text{coop}, \text{comp}\}$, we then compute a representative (type-level) weight vector by averaging over all encounters of that type:

$$\overline{\theta}^{\beta} = \left(\overline{w}_{I1}^{\beta}, \overline{w}_{I2}^{\beta}, \overline{w}_{G}^{\beta}\right) \tag{5}$$

which we treat as the "typical" sociality parameters of that HV type in subsequent simulations. The values of $\overline{\theta}^{\beta}$ are reported in Fig. 5.

(2) AV sociality design. For AVs, we do not calibrate weights from data. Instead, we directly design their sociality through an SVO angle φ_{AV}. We sweep φ_{AV} over the range $[-90^\circ, 90^\circ]$ and, for each angle, convert it into an AV weight vector using the same SVO-to-weight mapping as in the HV case.

$$\theta^{\mathrm{AV}}(\varphi_{\mathrm{AV}}) = \left(w_{I1}^{\mathrm{AV}}, w_{I2}^{\mathrm{AV}}, w_{G}^{\mathrm{AV}}\right) \tag{6}$$

(3) From simulations to payoff entries. When constructing the payoff matrix, we perform a closed-loop simulation for each strategy combination in Table 2 (for example, LT chooses AV while SG chooses a cooperative HV). In that simulation:

- if the player in a given role (LT or SG) is an HV of type β, its reward function uses the corresponding type-level weights $\overline{\theta}^{\beta}$;
- if the player in that role is an AV, its reward function uses the AV weights $\theta^{\mathrm{AV}}(\varphi_{\mathrm{AV}})$ for the AV sociality setting under study.

For each vehicle $i \in \{\text{LT}, \text{SG}\}$ in that run, we compute three trajectory-level feature components $R_{I1,i}$, $R_{I2,i}$, and $R_{G,i}$, corresponding to

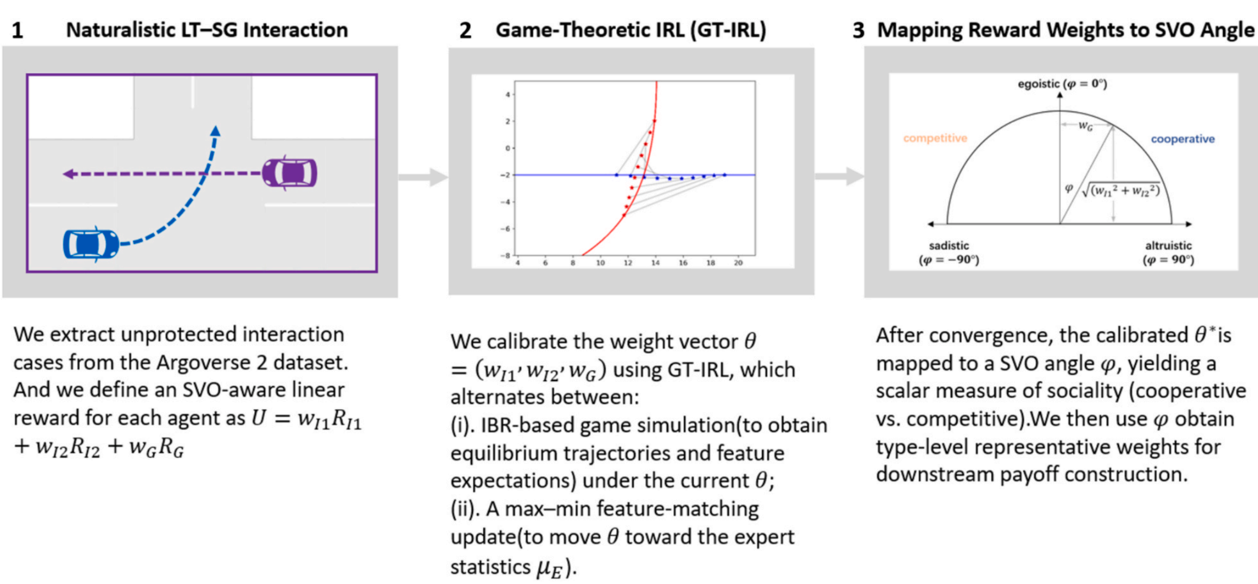

Fig. 4. Overview of the data-to-SVO angle pipeline: (1): Naturalistic LT–SG interaction extraction; (2): GT-IRL calibration of reward weights; (3): Mapping to SVO for payoff construction.

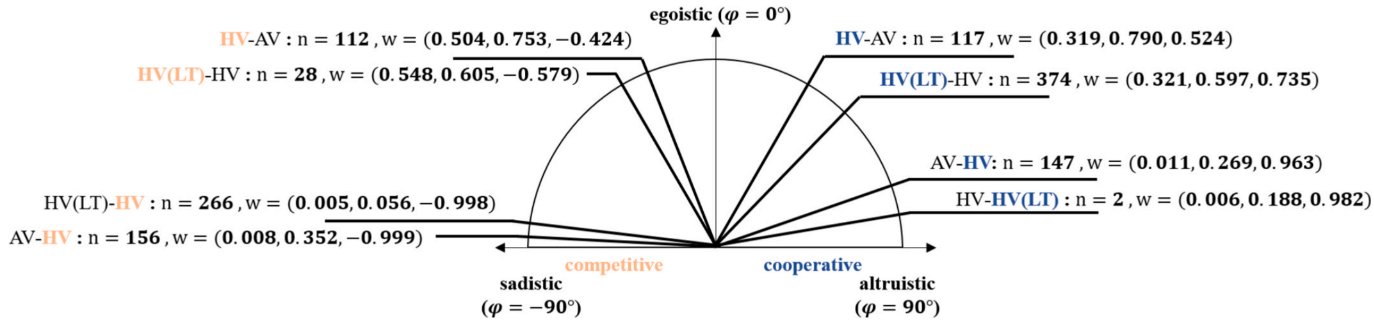

Fig. 5. Visualization of interaction cases and calibrated HV sociality weights in the SVO space (n denotes the number of cases, and $w = (w_{I1}, w_{I2}, w_G)$ denotes the calibrated reward weight vector).

Table 2
Payoff matrix of the left-turn vehicle group and the straight-going vehicle group.

	Strategy type	Straight-going vehicle HV_SG (p_{SG})		AV_SG ($1-p_{SG}$)	
Left-turning vehicle	HV_LT (p_{LT})	U_1^{LT}	U_1^{SG}	U_2^{LT}	U_2^{SG}
	AV_LT ($1-p_{LT}$)	U_3^{LT}	U_3^{SG}	U_4^{LT}	U_4^{SG}

driving progress, lane deviation, and conflict resolution over the entire interaction. The overall utility of vehicle i in that simulation is then obtained from the linear reward function in Eq. (1).

In our implementation, each strategy combination is simulated once, and the resulting trajectory-level utilities of the LT and SG vehicles, U_{LT} and U_{SG}, are directly used as the payoff entries in the corresponding cell of Table 2.

This role-independent random matching follows the standard large-population setting in evolutionary game theory and population games (Weibull 1997, Shulman 2012), where agents are repeatedly and anonymously paired to play a stage game and population states evolve according to average payoffs. In our naturalistic dataset, LT–SG encounters are short and spatially dispersed, so the probability that the same two drivers meet again is very low, making random matching a reasonable first-order approximation for the population-level analysis.

By the replicator dynamics, the HV growth rate in LT equals its excess payoff over the population mean, as shown in Eq. (7); analogously for SG, which is given by Eq. (8).

$$F(p_{LT}) = \frac{dp_{LT}}{dt} = \left[U_2^{LT} - U_4^{LT} + \left(U_1^{LT} - U_2^{LT} - U_3^{LT} + U_4^{LT}\right) \times p_{SG}\right] \bullet p_{LT} \bullet (1 - p_{LT}) \tag{7}$$

$$F(p_{SG}) = \frac{dp_{SG}}{dt} = \left[U_2^{SG} - U_4^{SG} + \left(U_1^{SG} - U_2^{SG} - U_3^{SG} + U_4^{SG}\right) \times p_{LT}\right] \bullet p_{SG} \bullet (1 - p_{SG}) \tag{8}$$

Based on relevant research on two-population evolutionary games (Ying et al., 2018, Awaga et al., 2020), the properties of the Jacobian matrix (as shown in Eq. (9)) will determine the stability performance of the equilibrium points. In other words, the stability of the equilibrium points is related to the values of the determinant and the trace of the matrix. ESS only exist when $Det(J) > 0$ and $Tr(J) < 0$.

$$J = \begin{bmatrix} \dfrac{dF(x)}{dx} & \dfrac{dF(x)}{dy} \\ \dfrac{dF(y)}{x} & \dfrac{dF(y)}{y} \end{bmatrix} \tag{9}$$

Ritzberger and Weibull (Ritzberger and Weibull 1995), as well as Selten (Selten 1988), point out that in multi-populations evolutionary games, strict Nash equilibria are necessarily pure strategies, and the stable solutions in such games correspond to strict Nash equilibria—namely, pure strategy profiles. Thus, our model admits up to four corner equilibria, whose determinants and traces are listed in Table 3; once the payoff matrix is specified, this table can be used to assess whether each equilibrium satisfies the ESS conditions.

4. Experiment results and discussion

We first calibrate SVOs for AVs/HVs and construct payoff matrices; we then compute ESS and analyze long-run dominance across sociality preferences. Using these calibrated preferences, we simulate AV–HV interactions to obtain the corresponding game payoff matrices. Based on the model derived in Section 3, we then compute ESS and analyze which AV sociality preferences yield a long-run advantage. Finally, by exploring the adaptation and evolution of strategies over longer horizons, we summarize design and policy implications for autonomous driving.

4.1. Data collection

Argoverse 2 Dataset: The Argoverse 2 dataset is collected by AVs under Argo AI on actual roads across six U.S. cities (Wilson, Qi et al.). It comprises 1,202 instances of unprotected left-turn scenario. Each case in the dataset has a total duration of 11 s and provides essential information, including scenarios ID, trajectory recording details, and map data.

After segmenting full trajectories and identifying the interaction segments, we analyze these instances—comprising 670 HV-HV, 303 AV (LT)-HV, and 229 HV (LT)-AV instances. Due to the lack of AV-AV data, these instances have been substituted with AV-HV interactions. Furthermore, since AVs exhibit similar decision-making processes in both AV-HV and AV-AV interactions, they can be effectively modeled using the SVO framework, making this substitution method technically valid.

We then apply the SVO calibration method outlined in Section 3 to all cases, thereby obtaining weights and sociality preferences that ground the payoff construction.

Recognizing that human drivers are heterogeneous (exhibiting a range of cooperative vs. competitive tendencies), we cluster HVs into cooperative and competitive. This heterogeneity must be accounted for in our modeling process. By analyzing HVs' data from the natural driving dataset, we can classify HVs into competitive types and cooperative types, as shown in Fig. 5 summarizes the resulting HV types. For

Table 3
The determinant and trace of the Jacobian matrix based on the equilibrium points.

Equilibrium Points	*Det(J)*	*Tr(J)*
(0, 0)	$(U_1^{LT} - U_2^{LT})*(U_3^{SG} - U_4^{SG})$	$U_2^{LT} - U_4^{LT} + U_3^{SG} - U_4^{SG}$
(0, 1)	$(U_1^{LT} - U_3^{LT})*(U_3^{SG} - U_4^{SG})$	$U_1^{LT} - U_3^{LT} + U_3^{SG} - U_4^{SG}$
(1, 0)	$(U_2^{LT} - U_4^{LT})*(U_1^{SG} - U_2^{SG})$	$U_4^{LT} - U_2^{LT} + U_1^{SG} - U_2^{SG}$
(1, 1)	$(U_1^{LT} - U_3^{LT}*(U_1^{SG} -) U_2^{SG})$	$U_1^{LT} - U_3^{LT} + U_1^{SG} - U_2^{SG}$

each interaction type and HV sociality preference, we report the number of cases and the mean calibrated weights (w_{I1}, w_{I2}, w_G). These statistics make the HV typing procedure transparent and reproducible. Based on this SVO-based classification, we construct two separate evolutionary game models between AVs and each HV type, in order to investigate which social preferences AVs should adopt when interacting with heterogeneous human drivers.

4.2. Data and calibration pipeline

We develop a closed-loop pipeline that maps real trajectories data to evolutionary outcomes through the following steps:

(1) Case extraction. Extract interaction cases from unprotected left-turn scenarios in the Argoverse 2 dataset, forming the event set for this study.
(2) IRL-based calibration. Using the reward structure in Section 3, we apply inverse reinforcement learning to each case to estimate the three normalized weights w_{I1},w_{I2} and w_G. These weights are then mapped to a sociality preference angle φ via the defined projection.
(3) Population typing. For HVs, we aggregate per-case weights/angles and cluster into two behavioral types (cooperative vs. competitive). Each HV type is represented by the type-wise mean weights. For AVs, we likewise compute representative cooperative/competitive settings by averaging across relevant cases.
(4) Payoff construction. Given LT/SG roles and an AV/HV pairing, we compute encounter payoffs from the calibrated (SVO-informed) reward, producing two 2×2 payoff tables (one for LT, one for SG).
(5) Angle sweep. To examine the sensitivity of evolutionary outcomes to AV sociality, we sweep $\varphi \in [-90^\circ, 90^\circ]$. For each angle, we calculate the corresponding weights, recompute the payoffs for all strategy pairings (as in Step 4), and form the corresponding payoff matrices.
(6) Evolutionary analysis. For each payoff specification, we solve replicator dynamics, determine ESS via the Jacobian test, and visualize evolutionary trend.
(7) Individual-based validation. Finally, we run finite-population, discrete-time Fermi imitation simulations (random LT–SG matching) to verify trajectories and end states under bounded rationality.

This pipeline links real interaction data → calibrated social preferences → payoffs → population dynamics, ensuring that strategic conclusions reflect observed preferences and their uncertainty.

4.3. Analysis of stable equilibrium solutions and sensitivity assessment

In this section, we analyze the ESS emerging from the evolutionary games constructed with the calibrated preferences. In this context, AV and HV denote strategies in each population. We use AV(LT) $=$ 1 (resp. AV(SG) $=$ 1) to indicate that the AV strategy is the evolutionarily stable outcome (dominant in the long run) in the LT (resp. SG) population; AV $=$ 0 analogously indicates that HV dominates due to higher relative fitness. When AV(LT) $=$ 1 and AV(SG) $=$ 1 simultaneously hold, the corresponding AV sociality preference ultimately dominates both populations in the long-run dynamics.

Referring to Fig. 6 (a), for cooperative HVs, AVs with relatively competitive sociality preferences sustain dominance across both LT and SG (e.g., $\varphi \in [-87^\circ, -51^\circ]$). By contrast, strongly cooperative AV preferences (e.g., $\varphi \in [61^\circ, 90^\circ]$) gradually lose presence in both populations under extended interaction, consistent with the ESS outcomes.

As shown in Fig. 6 (b), when AVs interact with competitive HVs, AV strategies dominate in both populations when the SVO angle φ lies in a moderately egoistic range (e.g., $\varphi \in [-20^\circ, 9^\circ]$). In contrast, extreme orientations—either highly competitive or highly cooperative (e.g., $\varphi \in [-90^\circ, -86^\circ]$ and $\varphi \in [71^\circ, 90^\circ]$—fail to persist in the long run.

4.4. The impact of HV social preference on the evolution of AV

Building on the previous analysis, the optimal sociality preference for AVs depends on the social preferences of the HVs they interact with. To further explore how AVs co-evolve with both competitive and cooperative HVs, we select AV strategies from the optimal social preference intervals derived in Section 4.2, and visualizes their evolutionary dynamics through phase diagrams under different HV sociality preferences.

As shown in Fig. 7 (a), when interacting with competitive HVs, both the LT and SG populations converge monotonically toward the AV strategy. Convergence is fastest when the AV share in the population is near 0.5, reflecting larger payoff differentials in that region; conversely, evolution slows when AV is either rare or already dominant (AV share near 0 or 1), where payoff gradients are small.

In contrast, Fig. 7 (b) shows that in the context of interactions with cooperative HVs, the evolutionary trajectory of AVs does not move directly toward AV $=$ 1, but instead follows a distinct turning path: the system initially deviates from the main diagonal, then bends back and ultimately converges to the globally stable state at (1, 1). This detour arises from saddle points near (0, 0) and (0, 1) that exhibit local attraction along certain directions but are not long-run stable. The resulting paths reveal a temporal asymmetry between LT and SG: one population may adopt AV earlier while the other lags, consistent with role-specific interaction intensity (LT faces tighter gaps and more complex timing at unprotected intersections).

In summary, AVs exhibit distinct evolutionary patterns when interacting with different HVs. In competitive environments, AVs tend to adapt quickly and gain dominance, resulting in faster convergence. Conversely, in cooperative settings, the interaction dynamics evolve more slowly. This phenomenon has practical implications for real-world

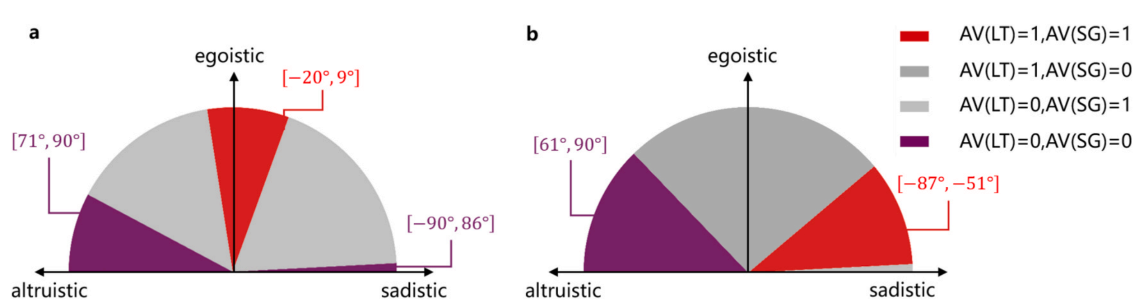

Fig. 6. Distribution of ESSs. Red regions indicate sociality preferences where AVs dominate; purple regions indicate HV dominance. (a) With competitive HVs, AVs succeed with mild egoism. (b) With cooperative HVs, AVs succeed with competitive orientation. The map reveals a "tough-on-soft, soft-on-tough" strategy: AVs could achieve dominance by being strategically assertive towards cooperative HVs, yet strategically accommodating towards competitive HVs to avoid mutually detrimental conflicts. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

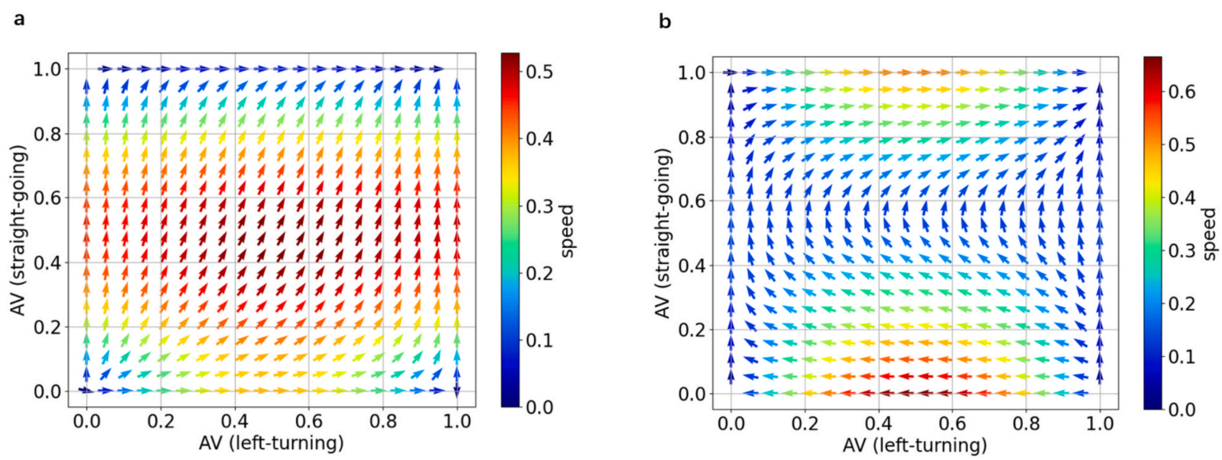

Fig. 7. (a) Evolutionary trend of AVs interacting with HVs exhibiting competitive preferences. (b) Evolutionary trajectories of AVs interacting with HVs exhibiting cooperative preferences. In both figures, arrows indicate the direction of evolution among different strategies, and arrow colors represent the speed of strategy change. The speed is calculated as the magnitude of the vector field derived from the replicator dynamic equations, that is, the Euclidean norm of the derivative vector at each strategy combination point. All speed values are normalized for visualization purposes.

AV deployment: for example, developing countries such as China are characterized by more competitive HV behavior, which may accelerate the evolution of certain AV strategies; whereas in the U.S. and EU, where HVs tend to be more conservative (Liu et al., 2024), AV strategies evolution may proceed more gradually.

4.5. The impact of varying penetration rates on the evolution of AVs

As suggested above, initial market penetration influences how quickly AV strategies spread. We therefore examine evolutionary dynamics under different initial AV shares.

In Fig. 8 (a), even with low initial AV penetration, both populations ultimately converge to AV dominance. This robustness indicates that, against competitive HVs, the AV strategy maintains a persistent payoff advantage, and the basins of attraction favor AV in the long run.

In contrast, Fig. 8 (b) illustrates the dynamics when AVs interact with cooperative HVs. When the initial market share of AVs is below 0.5, the LT population's AV share may decline initially before crossing a turning threshold and then rising toward dominance. Intuitively, a higher initial AV share increases the chance of AV–AV encounters, which yield higher payoffs to AVs, accelerating growth; when HV–HV interactions dominate, redundant or inefficient cooperation can depress system payoffs, eventually enabling AVs to regain momentum and expand.

This finding mirrors real-world observations where human drivers can temporarily gain an advantage by exhibiting aggressive behaviors, such as cutting in or abrupt lane changes, particularly when interacting with overly conservative AVs. Such interactions may reduce public confidence in AVs and hinder their societal acceptance, potentially leading to a decline in AV deployment. However, as the frequency of HV-HV interactions increases, competitive tendencies intensify, lowering the quality of interactions and system efficiency. This collective effect incentivizes a renewed increase in AV adoption, allowing AVs to regain momentum and move towards steady growth.

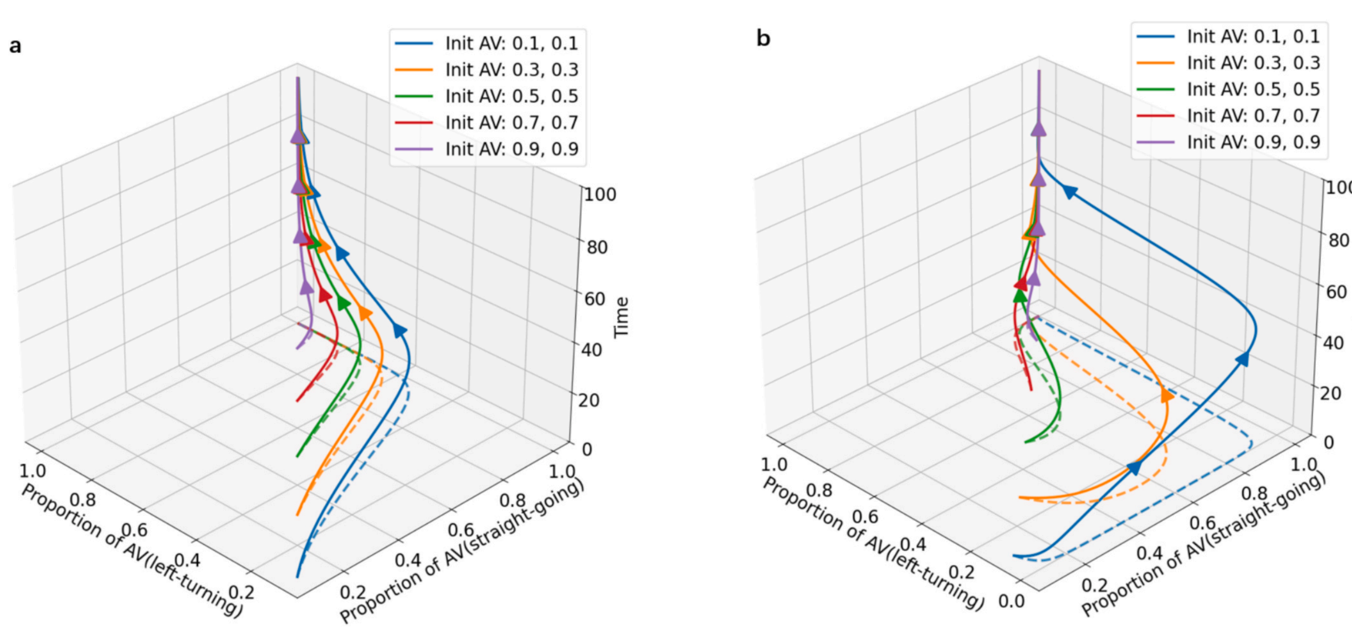

Fig. 8. (a) Different evolutionary trajectories of AVs (competitive type) under different initial penetration rates. (b) Different evolutionary trajectories of AVs (cooperative type) under different initial penetration rates. In both figures, the different curves illustrate how varying initial penetration rates of AVs affect their evolutionary process. The times axis denotes the number of times the evolutionary game is played.

4.6. Individual-based evolutionary simulation

To address the limitations of replicator dynamics under the continuous-time, infinite-population assumption and to further verify that our conclusions generalize to real-world interactions, we build an individual-based evolutionary simulation with Fermi rule (Szabó and Tőke 1998, Traulsen et al., 2006). The Fermi rule is an update rule in which an agent compares its payoff to another's and probabilistically adopts the other's strategy, with the probability given by a logistic function of the payoff difference. The simulation uses finite population size, discrete time, and random matching to approximate real-world settings.

We consider an unprotected left-turn scenario with two populations: LT and SG, each initialized with $N = 100$ agents. Each agent adopts one of three strategies: AV, HV1 (competitive), or HV2 (cooperative). The payoff computation and updating rules are as follows:

- Payoff computation. Each simulated encounter draws payoffs from the Section 4 calibrated 2×2 payoff tables (separate tables for LT and SG roles). To capture opponent-specific adaptation, the AV uses opponent-specific SVO angles (e.g., -64° vs. HV1 and -4° vs. HV2) selected from the optimal interval. Given the opponent and angles, the corresponding entries are read from the appropriate table and used in the simulation.
- Random matching and updating. At each discrete step, we uniformly sample one LT and one SG agent to play (asynchronous, one match per step). Strategy revision is permitted only along feasible pairs (AV–HV1, AV–HV2, HV1–HV2); same-type pairs (e.g., AV–AV, HV1–HV1, HV2–HV2) do not trigger updates and are skipped. Adoption follows the Fermi rule:

$$p = \frac{1}{1 + e^{-\beta \bullet \Delta}} \tag{10}$$

where β denotes the selection intensity (larger β means choices respond more strongly to payoff differences), and Δ is the payoff difference, computed as "candidate payoff − current payoff" against the same opponent.

The simulation terminates once the step limit is reached or when either the AV or HV share reaches 1. For each run, we track the time series of the three strategy shares and report averages across multiple random seeds, with optional confidence bands. Under finite populations, discrete time, random matching, and bounded rationality, the individual-based evolutionary dynamics align with the replicator-equation analysis in both trajectory and terminal state.

As shown in Fig. 9, (a) implements an opponent-aware policy: the AV adapts its SVO by opponent type ($\varphi = -4^\circ$ vs. HV1; $\varphi = -64^\circ$ vs. HV2), and the AV share in both LT and SG populations rises monotonically toward fixation (HV2 disappears first, then HV1). (b) and (c) each adopt a fixed SVO chosen from the analytically preferred angles ($\varphi = -4^\circ$ or $\varphi = -64^\circ$, respectively) but without conditioning on the opponent. In both cases the AV goes extinct (faster under $\varphi = -4^\circ$), despite each angle being optimal against one HV type in isolation. The contrast between (a) and (b)/(c) demonstrates that opponent-aware, dynamically switching sociality preference—not static, single-angle optimization—-drives population-level diffusion and resolves the social dilemma in mixed traffic.

4.7. Summary and insights derivation

This section integrates the experimental analyses and yields four insights:

(1) Dynamic identification and adaptation. AVs must adapt SVO online to heterogeneous HV preferences; static, one-size-fits-all sociality underperforms across populations.
(2) Avoid extreme cooperativeness. Overly cooperative AV policies are evolutionarily unstable and tend to be eliminated over time, while non-extreme, balanced self–other trade-offs deliver robust long-run advantages under both HV regimes.
(3) Context matters. The evolution of AV dominance depends on the prevalent HV preference distribution (competitive vs. cooperative mix); design and policy should be population-aware.
(4) Deployment is a marathon. At low penetration, AV advantages can be masked by interaction noise; around mid-range (~50%), larger AV–AV exposure accelerates convergence. Engineering

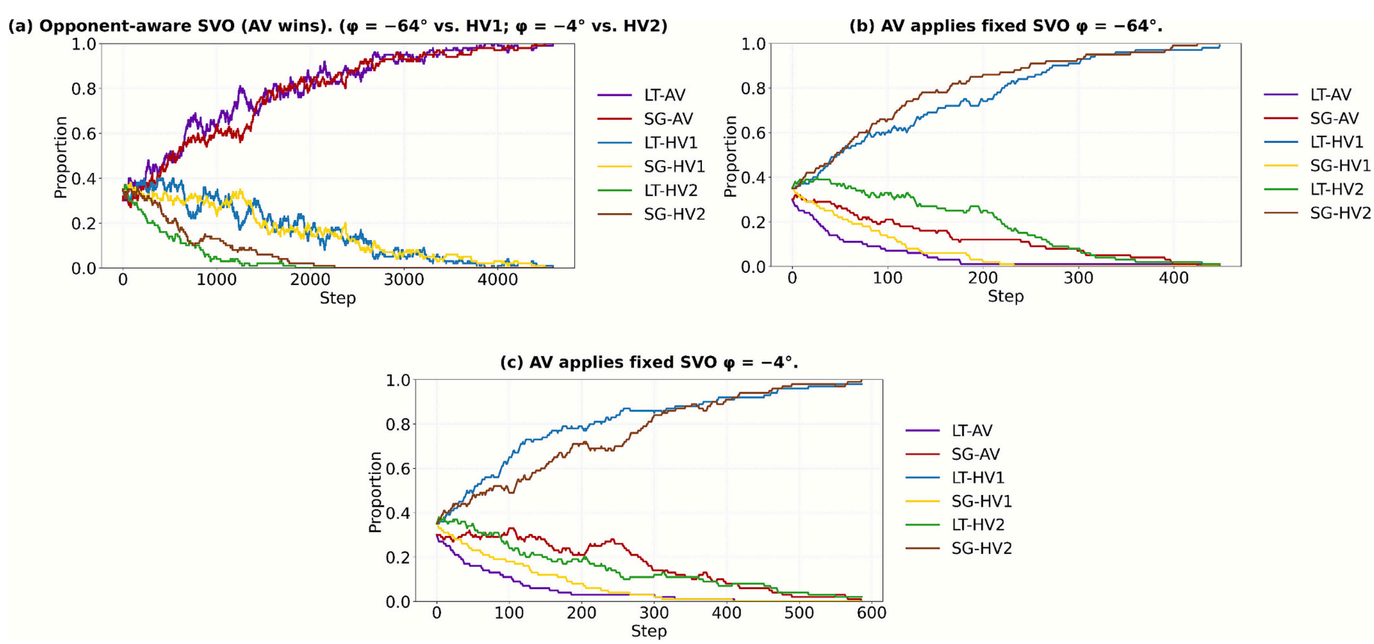

Fig. 9. Individual-based evolutionary simulation for an unprotected left-turn scenario. (a) AV succeeds with opponent-aware strategy. The AV population monotonically increases to fixation (LT: 1.00; SG: ≈0.99), sequentially eliminating HV2 and then HV1. (b) AV goes extinct with a strongly selfish SVO ($\varphi = -64^\circ$). The AV population is eliminated, and HV types expand to near fixation. (c) AV extinction is slower with a moderately egoistic SVO ($\varphi = -4^\circ$). Despite a slower decay rate, the AV still faces extinction, and HV types dominate.

and policy should therefore target long-horizon gains and support the system through the transient phase.

5. Conclusion

This study investigates the social dilemma faced by AVs in mixed traffic and proposes an evolutionary game-theoretic model that incorporates the SVO framework to analyze how AVs with different sociality preferences co-evolve with heterogeneous HVs. By characterizing equilibria via ESS, the model identifies AV sociality preferences that promote long-run adaptability and success. In particular, the analysis shows how appropriate sociality preferences help AVs escape social dilemmas and converge toward strategies better suited for sustained interaction—highlighting that while single encounters resemble sprints, population-level AV development is a marathon.

Based on the experimental analysis, we summarize three key strategic insights regarding the social behavior of AVs: (1) The optimal social preference for AVs varies depending on the type of HVs they interact with, calling for dynamic, opponent-aware adjustment; (2) Overly cooperative behavior is evolutionarily fragile; by contrast, moderately competitive orientations enhance AV adaptive advantage over time; (3) AV advantages are less apparent at low market penetration rates; as penetration approaches the mid-range (~50%), AVs' collective adaptive advantages become more salient.

Our analysis focuses on unprotected left-turn intersections with two strategies per role; generalization to multi-strategy settings and other scenarios (e.g., merging, ramps) remains to be validated. Additionally, the SVO framework used here serves only as a proxy for latent social preferences and lacks direct behavioral validation. The representation of HV heterogeneity is simplified and requires further refinement. Moreover, the evolutionary analysis relies on a role-independent random matching assumption between LT and SG drivers, which may underrepresent context-dependent or repeated interactions.

Future research can be expanded in several directions. First, incorporating additional evaluation dimensions—such as cost, comfort, and other user-centric factors—would enable a more comprehensive assessment of AV development potential. Second, evolutionary models should be extended to a broader range of scenarios—such as ramp merging and lane changing—to validate the generalizability of the study's conclusions. Furthermore, the observed asymmetric evolution pattern—where AVs dominate on one side while HVs dominate on the other—may stem from structural differences between left-turn and straight-going trajectories at intersections. This warrants further investigation into whether AVs with specific sociality preferences can exert structural influence on the behavioral patterns of opposing HVs.

CRediT authorship contribution statement

Li Rui: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Conceptualization. **Liu Yiru:** Methodology, Data curation. **Sun Jian:** Supervision, Resources. **Tian Ye:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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